

HTML Notes

A complete beginner-to-intermediate reference — compiled for class by Kapil.

01 · INTRODUCTION

What is HTML?

HTML is the combination of **HyperText** and **Markup Language**.

- **Hypertext** means machine-readable text — the browser only understands HTML.
- **Markup** means structuring text in a specific format. A markup language defines a text document using tags that describe the structure of a webpage.

HyperText Markup Language is the standard markup language for documents designed to be displayed in a web browser. It defines the **content and structure** of a webpage, and is often paired with:

- **HTML + CSS** — build the static webpage (structure + style).
 - **JavaScript** — adds dynamicity and functionality to the page.
-

02 · SETUP

Emmet abbreviation & the HTML boilerplate

Typing `html:5` then `Shift + ! + Enter` in an editor expands to the standard HTML5 skeleton:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
  </head>
  <body>

</body>
</html>
```

<!DOCTYPE html>

A required line that instructs the browser which version of HTML the page is written in. DOCTYPE stands for **Document Type Definition**.

In HTML5 it is a single simple line. Older documents (HTML 4 / XHTML) needed a full reference to a DTD:

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN"
"http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">
```

<html lang="en">

lang="en" tells the browser the page's language is English.

<head> ... </head>

The container for information *about* the document (not shown directly on the page). It sits between the <html> and <body> tags.

The <meta> tag

Lives inside <head> and describes "data about the data" — who wrote the page, its summary, character set, keywords, description, refresh rate, and viewport.

```
<meta charset="UTF-8">
```

Defines the character set. The Unicode Consortium maintains the Unicode Standard, whose goal is to replace older character sets with one universal **Unicode Transformation Format (UTF)**.

Note: Unicode is a character set — a list of characters mapped to unique code points (A = 65, B = 66, C = 67 ...). UTF-8 is an *encoding* of that set.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

- **Viewport** — the user's visible area of the page.
- **width=device-width** — matches the screen width of the device.
- **initial-scale=1.0** — sets the initial zoom level on load.

03 · FUNDAMENTALS

Tags & elements

An HTML document is built using **tags** — text wrapped between a left angle bracket and a right angle bracket, e.g. <tag-name>.

Two types of tags

Type	Description	Example
Paired (container)	Has an opening and a closing tag	<head></head>, <body></body>
Unpaired (void / empty)	Has only an opening tag	<meta>

Elements

An **element** is a paired tag together with its content — an opening tag, the content, and the closing tag. Example: <h1>Hello</h1>.

Inline vs. block-level elements

Type	Behaviour
Inline-level	Occupies only the width of the element's own content
Block-level	Occupies the entire width of the line, from start to end

04 · CORE TAGS

Basic tags

Headings

Create headings in a document. All six — h1 to h6 — are paired, block-level tags.

Paragraph — `<p></p>`

Creates paragraphs. Only a single space is preserved between characters, and line breaks inside the tag are ignored.

Preformatted — `<pre></pre>`

Prints text exactly as written inside the tag — extra spaces and line breaks are preserved.

Div — `<div></div>`

Defines a division or section of a document. Block-level element.

Span — `<span></span>`

Marks up a part of the text or document. Inline-level element.

Image — `<img src="" alt="">`

- **src** — the source path of the image to embed.
- **alt** — alternate text shown on the page if the image can't be displayed.

Attribute: gives an HTML element extra or special information/state.

Path types

Type	Description
Absolute path	The complete address of a file — protocol (http:// / https://), domain name, and the full directory to the file.
Relative path	The location of a file relative to the current page — how many folders up/down and in which direction, from where the page sits.

05 · TEXT STYLING

Formatting tags

All formatting tags below are **inline-level** elements:

`<b>`, `<strong>` · `<i>`, `<em>` · `<u>`, `<ins>` · `<mark>` · `<q>` · `<sub>` · `<sup>` · `<del>` · `<center>`

`<strike>` is **deprecated** — use CSS (`text-decoration: line-through`) instead.

Anchor tag

The anchor tag turns Hypertext into a **HyperLink** — but only when paired with the href attribute.

```
<a>HOME</a>           // plain Hypertext, not a link
<a href="...link...">HOME</a> // a real HyperLink
```

A paragraph with an href attribute — `<p href="...">HOME</p>` — does **not** create a link. The browser only recognises href on the anchor tag. To make a paragraph clickable, wrap it in an anchor:

```
<a href="..."><p>content</p></a>
```

Hypertext vs. HyperLink

HyperText	HyperLink
The visible text on the UI	Text content that links to another webpage
Created with any normal/basic tag	Created only with the anchor tag + href

Default link colours

State	Colour
Not visited	Blue
Visited	Purple
Active	Red

What an <a> tag can link

Content	Can link to
Text	Internal files
Image	A website
Button	A location

The target attribute

Value	Behaviour
_blank	Opens the hosted page in a new tab
_self	Opens the hosted page in the same tab
_top	Returns to the top of the page
_parent	Opens the page in the parent frame

Core attributes

"Core" as in the starting point — every HTML element can carry these four:

Attribute	Description
id	Must be unique — targets one individual element.
class	The same value can repeat — targets multiple elements at once.
title	Creates a tooltip — a small popup shown on hover over the element.
style	Inline CSS — styles the element directly.

08 · EMBEDDING

iFrames

An `<iframe>` runs a webpage inside another webpage. Height and width can be set on it, but not every website allows itself to be embedded this way — it's blocked for security reasons on many sites.

```
<iframe src="" frameborder="0" height="" width=""></iframe>
```

To embed a YouTube video: go to YouTube → open the video → Share → Embed → copy the embed code and paste it into your HTML.

09 · STRUCTURE

Semantic tags

Semantic tags describe their own meaning to both the browser and the developer — compare how a layout was built in HTML4 versus HTML5:

HTML version 4		HTML version 5	
<div id="header">		<header>	
<div id="nav">		<nav>	
<div id="section"> <div id="article">	<div id="aside">	<section> <article>	<aside>
<div id="footer">		<footer>	

Memory aid — "H D A M N S T A F F"

- H** Header
- D** Detail
- A** Article
- M** Main
- N** Nav
- S** Section
- T** Table
- F** Footer

- A** Aside
- F** Footer
- F** Figure, Figcaption, Form

10 · LISTS

Lists

HTML lists group related items together. There are three types:

1. Ordered list — `<ol>`

Each item is listed with `<li>`. Supported attributes:

Attribute	Description
type	"A", "a", "I", "i", "1" — default is Number
start	A number to start counting from, e.g. start="4"
reversed	Reverses the counting order

2. Unordered list — `<ul>`

Each item is listed with `<li>`.

Attribute	Description
type	"circle", "square", "disc", "none" — default is disc

3. Description list — `<dl>`

Uses two tags for each entry:

Tag	Meaning
<code><dt></code>	Description term
<code><dd></code>	Description data (appears with a small tab space by default)

11 · TABULAR DATA

Table

The `<table>` tag structures data into rows and columns.

Tag	Meaning
<code><tr></code>	Table row
<code><th></code>	Table heading
<code><td></code>	Table data

Table attributes

Attribute	Description
border="2"	Gives the table a border
cellpadding="15px"	Space between a cell's content and its border
cellspacing="10px"	Space between adjacent cell borders

colspan & rowspan

Attribute	Effect
colspan	Merges two or more columns into one cell
rowspan	Merges two or more rows into one cell

12 · USER INPUT

Forms

HTML forms collect data from the user.

- A form is built with the `<form>` tag — **block-level**, and it's semantic (every tag inside has meaning).
- A `<label>` tells the user what data to enter.
- An `<input>` creates the text box / field.
- All tags *inside* a form are inline-level.

```
<form>
  <label for="username">Name</label>
  <input type="text" name="username" id="username">
</form>
```

Key attributes

Attribute	Purpose
for / id	Links a <code><label></code> to its <code><input></code> — the values must match.
name	Acts as a container that holds the entered value (visible in the URL, e.g. <code>username=Raj</code>). Radio buttons and checkboxes in the same group must share the same name.
value	Required for radio, checkbox and select — without it, the submitted value defaults to "on".
placeholder	Shows faint hint text inside an empty input.

Form tags

- **Label** — paired tag; tells the user what to fill in.
- **Input** — paired tag; takes data from the user. Line breaks are not allowed inside it.

Input type	Description
text	Single-line text input

Input type	Description
password	Masked text input for passwords
checkbox	Toggle for selecting multiple options
radio	Single selection from multiple options
submit	Button to submit form data
button	General-purpose button
file	Upload files
number	Numerical values
date	Select a date
email	Email addresses
color	Select a colour
range	Slider for a value within a range
hidden	Hidden form data
image	Image used as the submit control

textarea

Takes data from the user — unlike `<input>`, line breaks *are* allowed. Paired tag.

- `cols="" / rows=""` — number of columns/rows shown.
- `style="resize:none"` — stops the user resizing it.

button

Paired tag used to create a button. Common types: `type="submit"`, `type="reset"`. Give the button visible text as its content.

fieldset & legend

- **fieldset** — paired tag; draws a border around a group of form fields.
- **legend** — paired tag; gives that border a name/caption.

Full example

```
<form>
  <fieldset>
    <legend>form</legend>
    <label for="username">Name</label>
    <input type="text" name="username" id="username">
    <button type="submit">Submit</button>
  </fieldset>
</form>
```


Audio tag

The <audio> tag embeds sound — music or any other audio stream.

```
<audio src="../../HTML/calling-21139.mp3" controls></audio>
```

Attribute	Value	Description
autoplay	autoplay	Starts playing as soon as it's ready
controls	controls	Shows audio controls (play/pause, etc.)
loop	loop	Starts over every time it finishes
muted	muted	Mutes the audio output

Video tag

The <video> tag embeds video content — a movie clip or other video stream.

```
<video src="../../HTML/istockphoto-1338832559-640_adpp_is.mp4" controls></video>
```

Attribute	Value	Description
autoplay	autoplay	Starts playing as soon as it's ready
controls	controls	Shows video controls (play/pause button, etc.)
height	pixels	Sets the height of the video player
loop	loop	Starts over every time it finishes
muted	muted	Mutes the audio output of the video